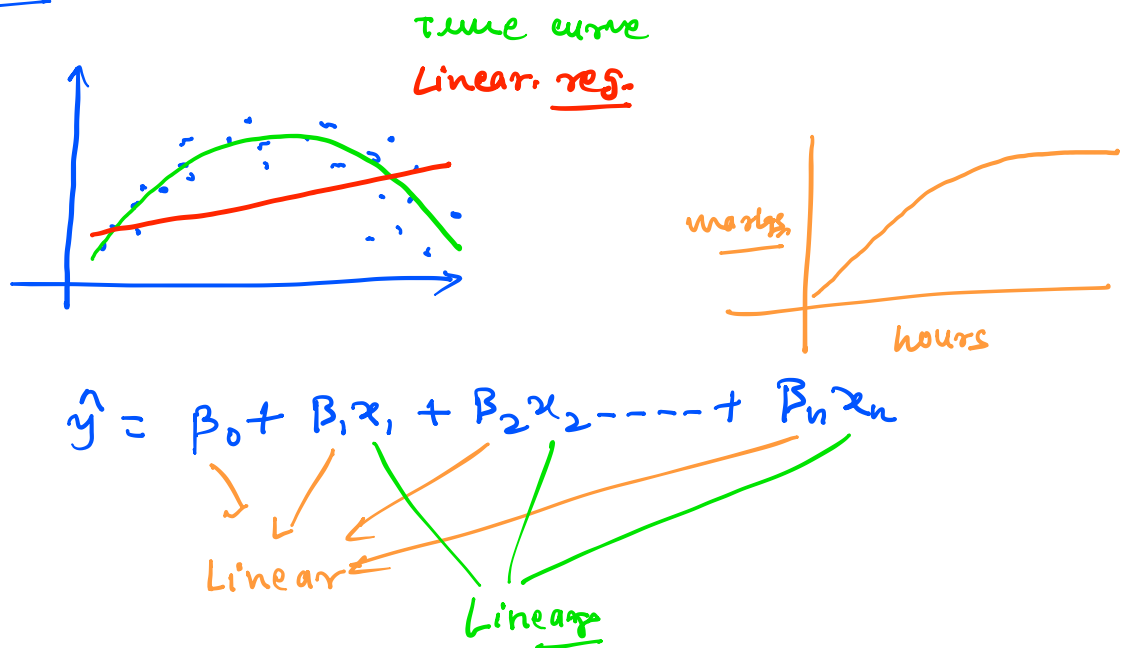


Polynomial Regression:

Linear reg.



polynomial regression:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 + \beta_4 x_2^2 + \beta_5 x_2^3 + \dots + \beta_d x_1^d$$

co-efficients are linear.

features are non-linear.

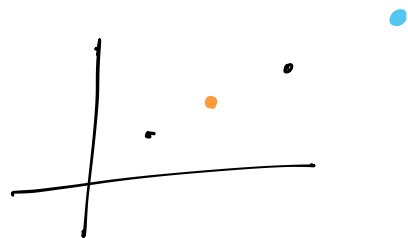
d defines the degree of the polynomial.

* curve fitting:

① interpolation

② smoothing,

③ least squares fitting.



$$y = f(x) + \epsilon$$

normal dist

$$f(x) = 3 + 2x - 0.7x^2, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$



Generalization:

The model performs well on training data as well as testing data.

good ML model $\neq$ lowest train error

good ML model = low future error
(unseen data)

{Train error}, {validation error, test error}

Error you get on training data.

Error measured on data not used for training.

Linear Algebra implementation:

$$X_{poly} = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_m & x_m^2 & x_m^3 \end{bmatrix}$$

$$\beta = [\beta_1 \ \beta_2 \ \beta_3 \ \dots \ \beta_d]^T$$

$$\hat{y} = X_{poly} \beta$$

$$x_1, x_2, x_1^2, x_2^2, \sqrt{x_2}$$

Bias and Variance!

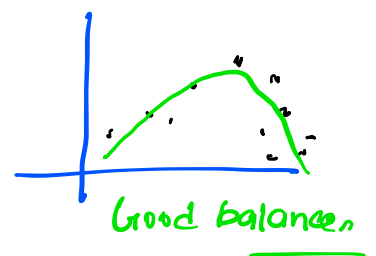
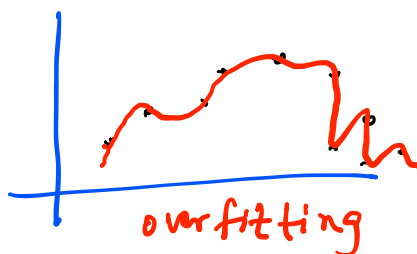
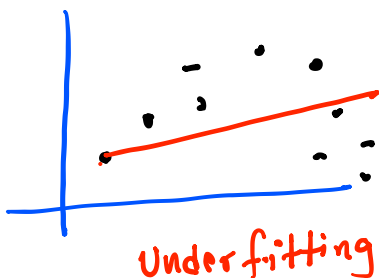
bias = error caused by the model being too simple.

how far the average prediction from the true pattern.

Variance! Error caused by model being too sensitive to the training set.

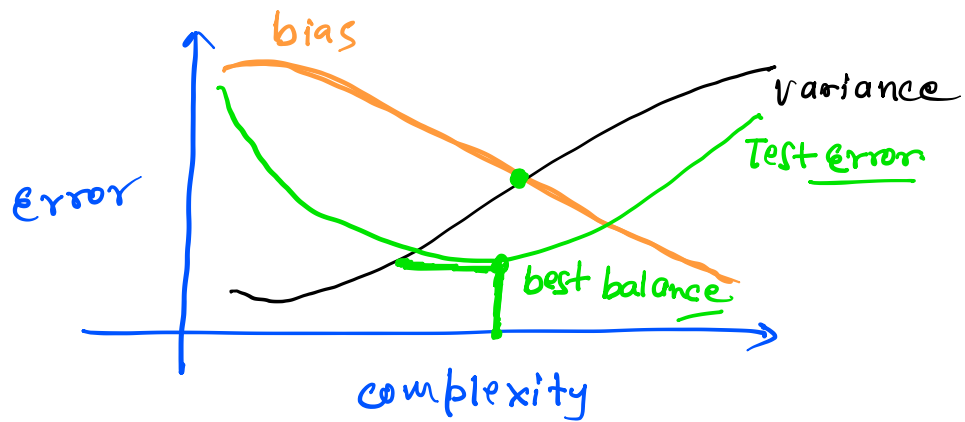
high variance = model changes a lot when training data changes

high bias $\rightarrow$ underfitting $\left\{ \begin{array}{l} \rightarrow \text{train error} \rightarrow \text{high} \\ \rightarrow \text{test error} \rightarrow \text{high} \end{array} \right.$
 high variance $\rightarrow$ overfitting. $\left\{ \begin{array}{l} \rightarrow \text{train error} \rightarrow \text{low} \\ \rightarrow \text{test error} \rightarrow \text{high} \end{array} \right.$

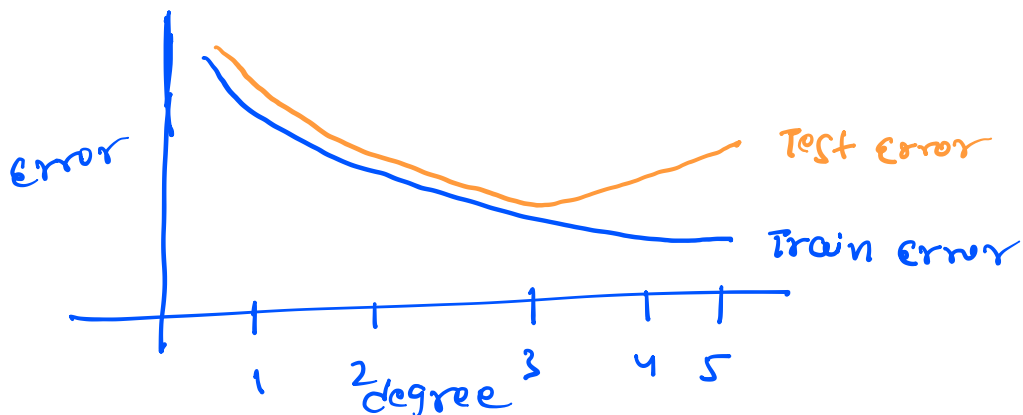
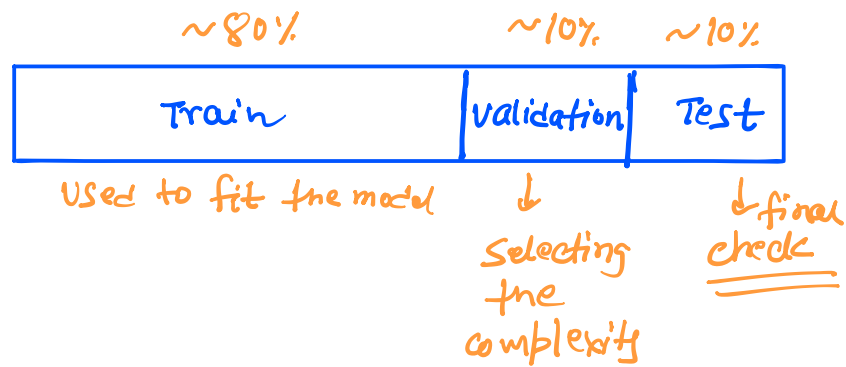


$$E[(y - \hat{y})^2] = \text{bias}^2 + \text{variance} + \text{noise}$$

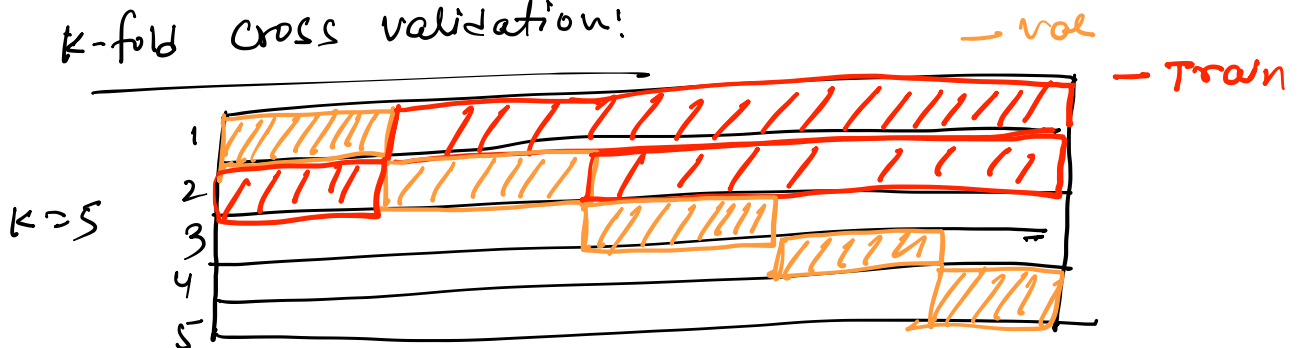
↓
Expectation



* Validation:



k-fold cross validation:



KFCV $\rightarrow$ Generalization.

$$d = \{1, 2, 3, 4, \dots, n\}$$

$$\text{selected degree} = \min_d \text{cv error}(d)$$

* Regularization:

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - (xw))^2$$

GT $\uparrow$ input $\rightarrow$ weight

objective $\min L$

$\min$ loss + penalty for complexity.

Ridge regularization:

$$L_{\text{ridge}}(w) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p w_j^2$$

$(0-1)$

power of the reg.

L_2 norm

Lasso regularization:

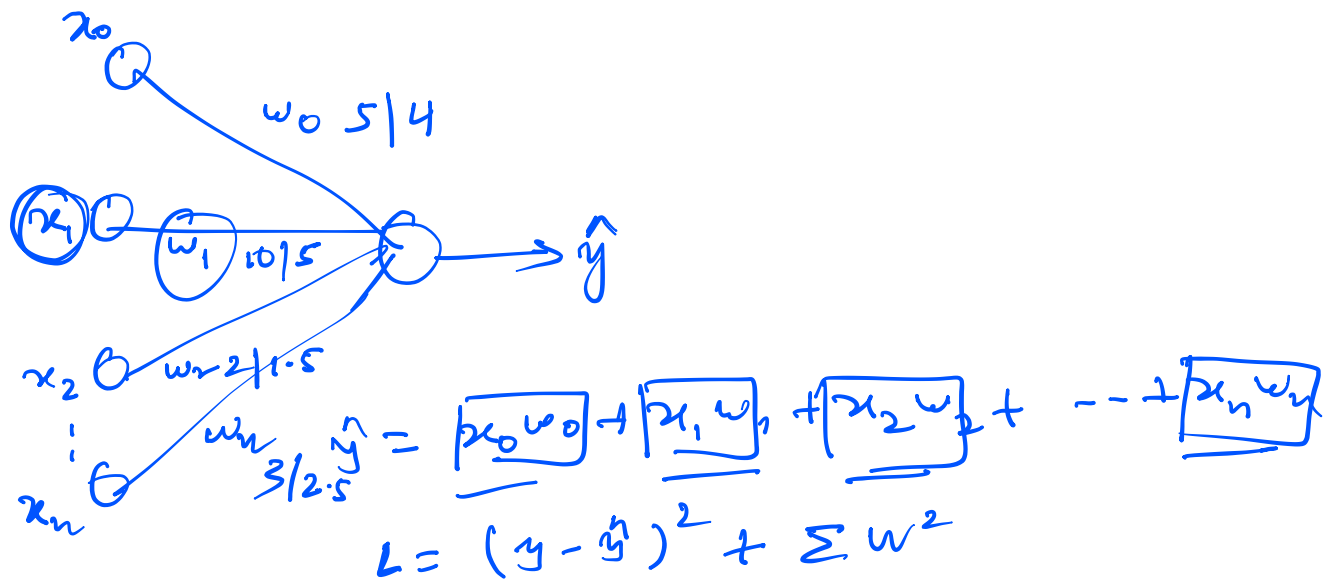
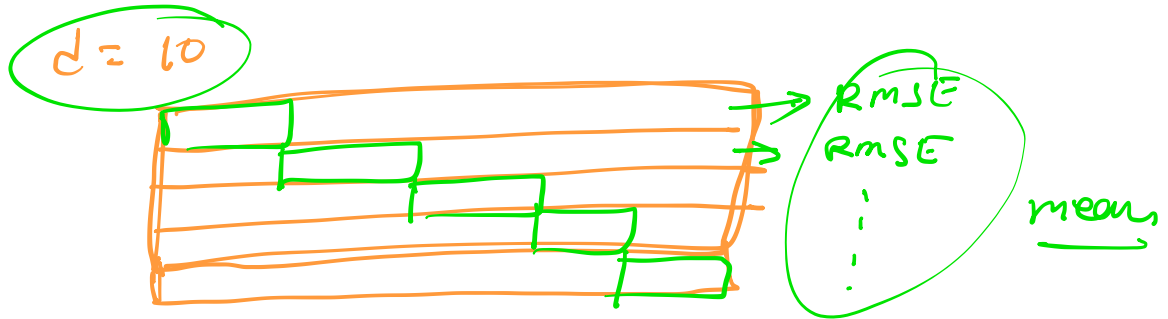
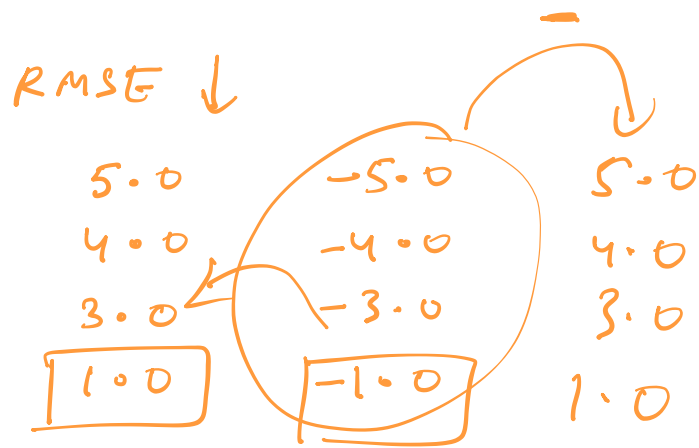
$$L_{\text{lasso}}(w) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |w_j|$$

L_1 norm

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$

0 $\mathbb{R}$ 0 $\mathbb{R}$



$$\frac{\partial L}{\partial w} =$$

$$w = \textcircled{w} - \frac{\partial L}{\partial w} \cdot \eta$$